

Essam Moussa Mahmoud

Full-Stack Developer – MERN Stack

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SUMMARY

Full-Stack Developer with hands-on MERN experience (MongoDB, Express.js, React.js, Node.js) building role-based, real-time, and API-driven web applications. Proven ability to design secure authentication systems (JWT, OAuth, bcrypt), model relational and document databases, and ship responsive React frontends with modern state-management patterns. Seeking a Junior Full-Stack or MERN Developer role to apply production-oriented engineering practices from day one.

Core Competencies: JavaScript (ES6+) · React.js · Node.js · Express.js · MongoDB · REST API Design · JWT Authentication · State Management (Redux, React Query) · Responsive Web Design · Git/GitHub

EDUCATION

Alexandria University

Computer and Data Science (In Progress)

Oct 2023 – Expected Jun 2027

TRAINING & PROFESSIONAL DEVELOPMENT

Digital Egypt Pioneers Initiative(DEPI) – Full-Stack Web Development **Nov 2025 – Jul 2026**

- Delivered full-stack applications using React and Node.js/Express within a structured, project-based curriculum, following sprint-style milestones.
- Architected reusable component libraries and implemented client-side routing and state management across multiple production-style applications.
- Collaborated in a Git/GitHub-based team workflow, applying code review, modular design, and version-control best practices.
- Designed and consumed REST APIs connected to SQL and NoSQL databases, implementing CRUD and business-logic operations end-to-end.
- Debugged and refined application logic under deadline pressure to deliver demo-ready, functioning features.

PROJECTS

EasyCare – Healthcare Appointment & Patient Management Platform [Code](#) | [Live Demo](#)

React.js · Node.js · Express.js · MongoDB · Mongoose · Socket.IO · JWT · bcrypt · Leaflet · Recharts

- Architected a full-stack MERN healthcare platform serving three distinct user roles (patient, doctor, admin) with scoped permissions and ownership checks enforced on every API route.
- Implemented secure multi-method authentication (JWT, bcrypt password hashing, Google OAuth) with role-based authorization middleware protecting all backend endpoints.
- Engineered a concurrency-safe booking system using MongoDB transactions to eliminate double-booking under simultaneous requests, and integrated Socket.IO for real-time patient-doctor chat.
- Built an admin analytics dashboard powered by MongoDB aggregation pipelines, enabling reporting, review moderation, and full appointment oversight, visualized with Recharts.
- Reduced [manual scheduling conflicts / support overhead] by [X]% through automated conflict detection at the database layer.

- Engineered a library management system handling book inventory, user accounts, and borrow/return workflows through a reusable, component-driven architecture.
- Implemented schema-validated forms with Zod and a secure authentication flow with sanitized input handling and structured error states.
- Separated global UI state (Redux) from server state (React Query) to enable real-time borrow/return updates without unnecessary re-fetching, improving perceived responsiveness.
- Structured the codebase around reusable, typed components to support long-term scalability and maintainability.

- Designed a responsive, modular admin dashboard using React and Material UI, prioritizing component reusability across views.
- Implemented interactive data visualizations and analytics using Nivo Charts to surface key metrics at a glance.
- Integrated calendar-based scheduling and event management with FullCalendar for streamlined operational workflows.
- Built validated, reusable form components with Formik and Yup, reducing invalid-input errors through client-side validation.

- Architected a full-stack MERN platform for discovering and booking wedding venues, backed by a REST API-driven Node.js/Express server.
- Designed MongoDB schemas for venues, availability windows, and booking records, exposing RESTful endpoints consumed by the React frontend.
- Built a responsive React UI for venue discovery, including detail pages with pricing, availability, and image galleries.
- Leading an in-progress UI/UX refactor to elevate the platform to a polished, production-grade visual standard.

CERTIFICATES

Certificate	Issuer	Year
Full-Stack Web Development	Digital Egypt Pioneers Initiative (DEPI)	2026
Developing Back-End Apps with Node.js and Express	Coursera IBM	2026
JavaScript Programming Essentials	Coursera IBM	2025
Developing Front-End Apps with React	Coursera IBM	2025
Getting Started with Git and GitHub	Coursera IBM	2025

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), HTML5, CSS3, SQL, Python, C++
- **Frontend:** React.js, Redux, React Query, Tailwind CSS, Material UI, Shadcn/UI, Responsive Web Design
- **Backend:** Node.js, Express.js, REST API Design, JWT Authentication, Socket.IO
- **Databases:** MongoDB, Mongoose, Database & Schema Design

- **Tools & Practices:** Git, GitHub, Postman, Agile/Scrum Fundamentals

LANGUAGES

- **Arabic – Native**
- **English – C1 (Advanced)**